

CRX Margin Framework Methodology

1. PURPOSE AND SCOPE

The margin framework sizes the collateral that protects the network against counterparty default. It defines three requirements per account: initial margin (IM), variation margin (VM), and maintenance margin (MM). Each requirement is computed continuously, netted at the account level, and enforced by the margin engine without discretion.

The framework recomputes each requirement on every mark update, and the update itself carries the full portfolio context: positions that offset one another reduce the requirement, and positions that concentrate risk raise it. The resulting number is a floor, not a target; an account may hold any balance above it.

2. RISK FACTOR MODEL

Every currency pair maps to a small set of risk factors. The factors are observable market quantities, and each position's exposure to them is linear over the margin horizon:

$$IM_p = ES_{97.5}(\Delta V_p, 2d)$$

The expected-shortfall calibration uses a 5-year window with stressed overlap, and the window rolls daily.¹ Factor correlations enter through the netting set, never per position.

Exhibit 1 — Risk factors by pair class

Pair class	Spot factor	Points factor	Basis factor	Horizon	Confidence
G10	Yes	Yes	No	2d	97.5%
EM deliverable	Yes	Yes	Yes	2d	97.5%
EM non-deliverable	Yes	Yes	Yes	2d	97.5%
Frontier	Yes	Yes	Yes	3d	99%

Horizon extends where local market closure risk is material.

¹ Calibration windows follow clearinghouse practice; see BIS, *Margin dynamics in centrally cleared markets*, bis.org.

3. NETTING AND AGGREGATION

Requirements net at the account level, never at the position level. The netting set is the account's full portfolio, and aggregation follows the factor model's correlation structure:

$$IM_{acct} = \sqrt{(\sum_{i,j} \rho_{ij} IM_i IM_j)}$$

Correlations floor at zero for margin purposes; negative correlation benefits accrue only through direct offsets within a factor.

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4. THE REQUIREMENT LIFECYCLE

Each requirement moves through a fixed lifecycle, and every transition is observable on-chain.

Exhibit 2 — Requirement lifecycle



- (a) a computed requirement becomes a call when it exceeds posted collateral;
- (b) a call unmet within its window opens a cure window;
- (c) a cure window unmet triggers closeout.

5. PARAMETERS

Parameters are published, versioned, and asymmetrically governed. CRX can tighten any parameter immediately. Loosening requires independent review.²

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² Review procedure [CONFIRM: REVIEWER IDENTITY]; current draft value [TBD].

6. WORKED EXAMPLE

A two-position EM portfolio illustrates the netting benefit. The example holds notional constant and varies only direction.

Exhibit 3 — Worked netting example

Portfolio	Gross IM	Net IM	Benefit
Long USD/BRL 1M + Long USD/KRW 1M	7.30%	6.15%	16%
Long USD/BRL 1M + Short USD/KRW 1M	7.30%	3.55%	51%
Long + Short USD/BRL 1M	7.30%	0%	100%

7. RELATIONSHIP TO COMPANION DOCUMENTS

This methodology sits beneath the Margin and Settlement Overview and beside the asset-level collateral terms. The overview states the rules; this document states the computation; the collateral overview states what satisfies the resulting requirement.³

³ CRX, *Margin and Settlement Overview* (July 2026); CRX, *Collateral Overview* (July 2026).